

Multiple Choice (10 marks)

1. How can one demonstrate that amino acids are used to build proteins in cells, while the proteins already in the cell are used in catabolism?
 - (a) By observing the cell under a microscope
 - (b) Using fluorescent tagging of proteins
 - (c) Tracking changes in cell size and shape
 - (d) Autoradiography with labelled amino acids
 - (e) Through protein synthesis
2. Which of the following is a biotic factor that could affect the growth rate of a population?
 - (a) Flash flood
 - (b) Tsunami
 - (c) Sudden reduction in the animal food resource
 - (d) Destruction of the ozone layer
 - (e) Drought
3. What is an oncogene?
 - (a) Oncogenes are found only in viruses
 - (b) Oncogenes prevent cancer cells
 - (c) Oncogenes are a type of immune cell
 - (d) Oncogenes detoxify harmful substances in the cell
 - (e) Oncogenes slow down cell division
4. To prevent desiccation and injury, the embryos of terrestrial vertebrates are encased within a fluid secreted by the
 - (a) blastoderm
 - (b) mesoderm
 - (c) amnion
 - (d) zona pellucida
 - (e) ectoderm
5. This biome contains plants whose roots cannot go deep due to the presence of a permafrost.

- (a) Savanna
- (b) Taiga
- (c) Tundra
- (d) Chaparral
- (e) Alpine

True / False (10 marks)

Answer True or False

- 6. True or False: Implementation of epidural analgesia for labor: is the standard of effective analgesia reachable in all women?
- 7. True or False: Are intraoperative precursor events associated with postoperative major adverse events?
- 8. True or False: Do inhaled steroids differ from cromones in terms of hospital admission rates for asthma in children?
- 9. True or False: Spinal subdural hematoma: a sequela of a ruptured intracranial aneurysm?
- 10. True or False: Transgastric endoscopic splenectomy: is it possible?

Long Form Response (20 marks)

Answer each question with a clear and complete explanation.

- 11. Discuss the key modifications that occur during RNA processing of preRNA, including the roles of point mutation repair, intron removal, and the addition of the 5 cap and poly-A tail.
- 12. Discuss the observations made by the biology student regarding the filamentous appendages on the bacterium. Explain why the conclusions about flagella and non-motility are incorrect, and identify the true nature of these appendages.

End of Paper